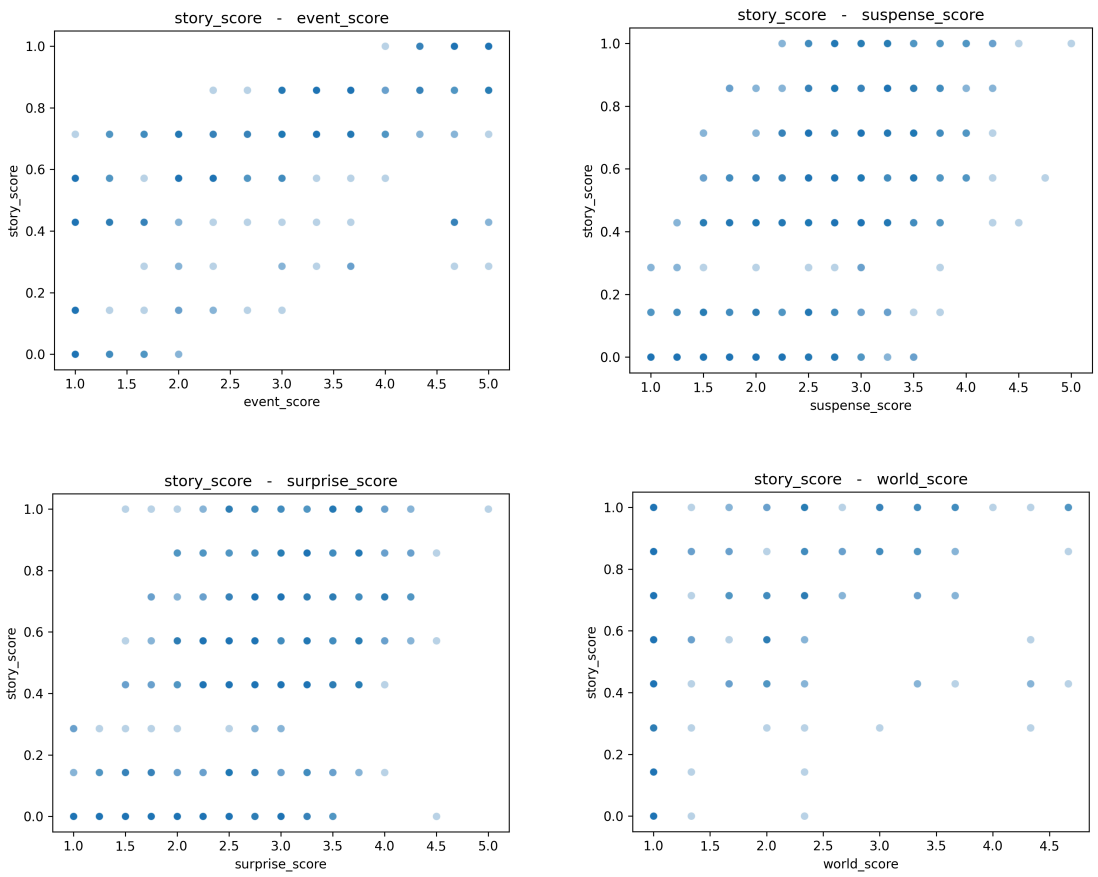
Assignment 2. Simple Linear Regression

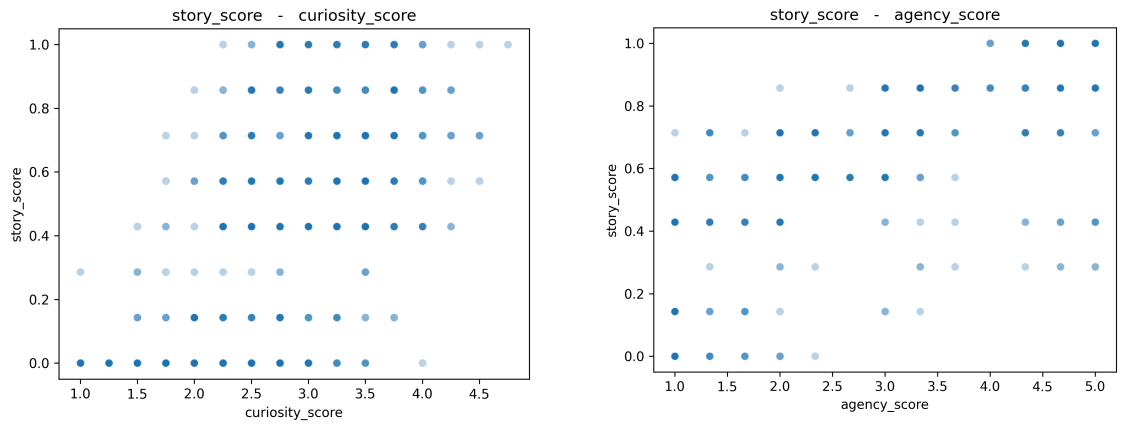
9 March 2026, [GitHub Code]

This report discusses the results of the application of multiple instances of simple linear regression, where the *Story Score* (0-1 scale) is the dependent variable and the independent variable changes between *Agency*, *Event Sequencing*, *World Building*, *Curiosity*, *Surprise*, and *Suspense.* The goal is to find out whether any of the independent variables is a good predictor for the *Story Score.*

The scatterplots (Figure 1.) give a first impression of how the dependent in independent variables compare. In these plots, darker dots means several dots are overlapping and therefore more frequent. The dots in the plots for *Curiosity*, *Suspense* and *Surprise* seem to be grouping in the bottom left to the top right of the graph, suggesting the possibility of some correlation. The graph of *World Building* (world_score) seems to group in the top left, indicating a low *World Building* score makes for a high *Story Score*. This world_score scatterplot seems the least 'linear'.

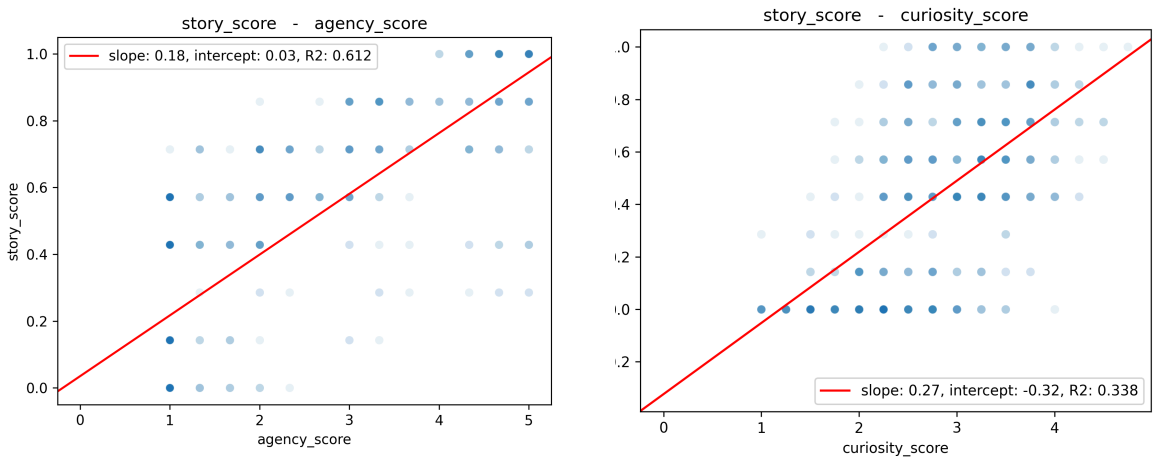
Figure 1. *Scatterplots of Story Score vs Various Independent Variables*

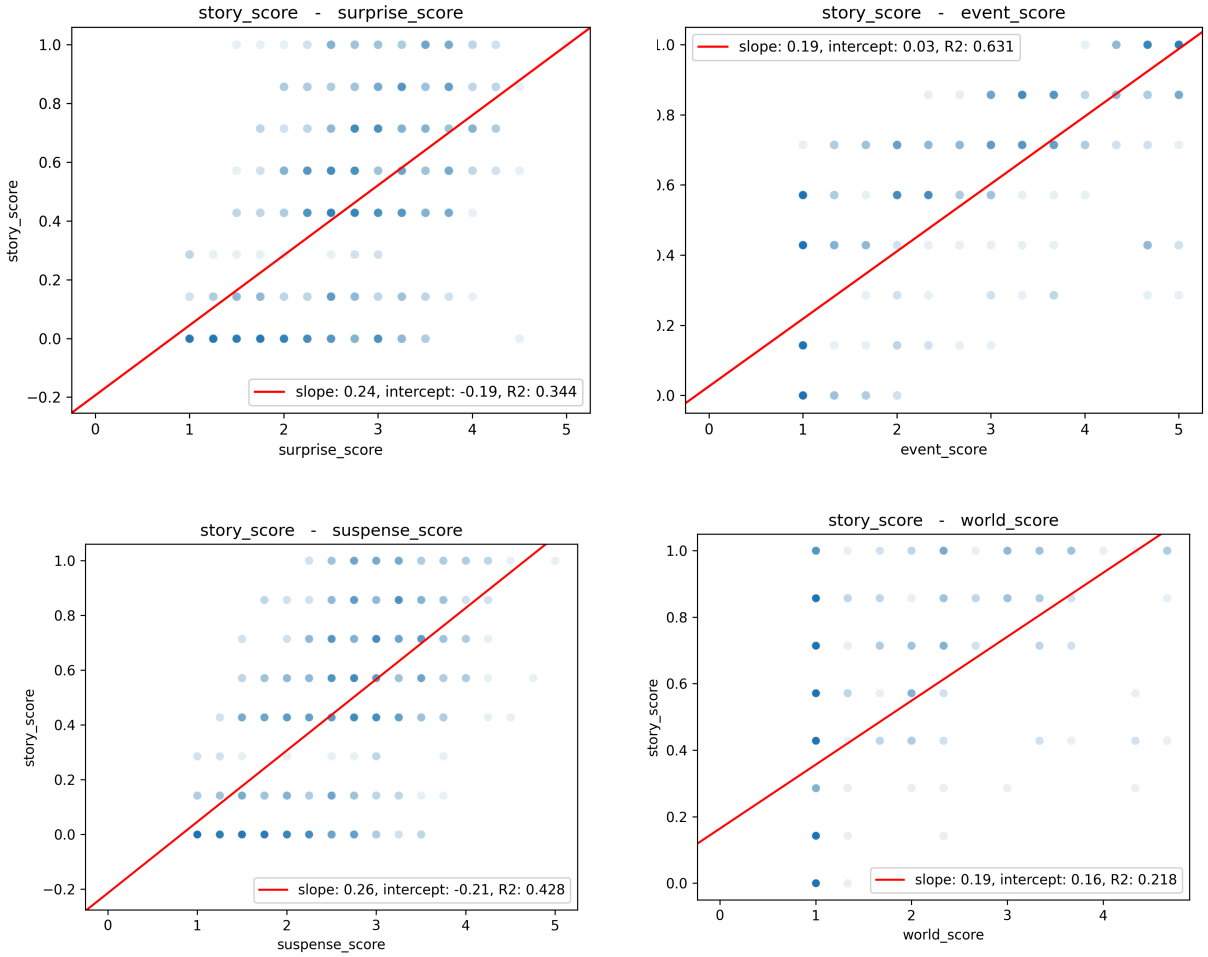




A more statistical approach to determine how well each independent variable predicts the *Story Score* is linear regression and calculating the coefficient of determination (R^2). We plot the regression line (Figure 2.) using the slope (angle) and intercept (value on which the regression line crosses the y-axis). These values tell us the optimal regression line that minimizes the sum of the squared errors of the residuals; or in simpler terms the best 'fit' for the data.

Figure 2. *Scatterplots with Regression Lines, R2, Slope and Intercept*





The independent variables *Agency* and *Event Sequencing* have the highest coefficient of determination (R^2), and in line with the guess from seeing only the scatterplots, *World Building* has the lowest R^2 (Table 1.). *Event Sequencing* has a R^2 of 0.63, which means that 63% of the variance (from the mean) of the dependent variable is explained by this independent variable. In simpler terms, out of all the independent variables, *Agency* and *Event Sequencing* appear to be the best predictors for *Story Score*.

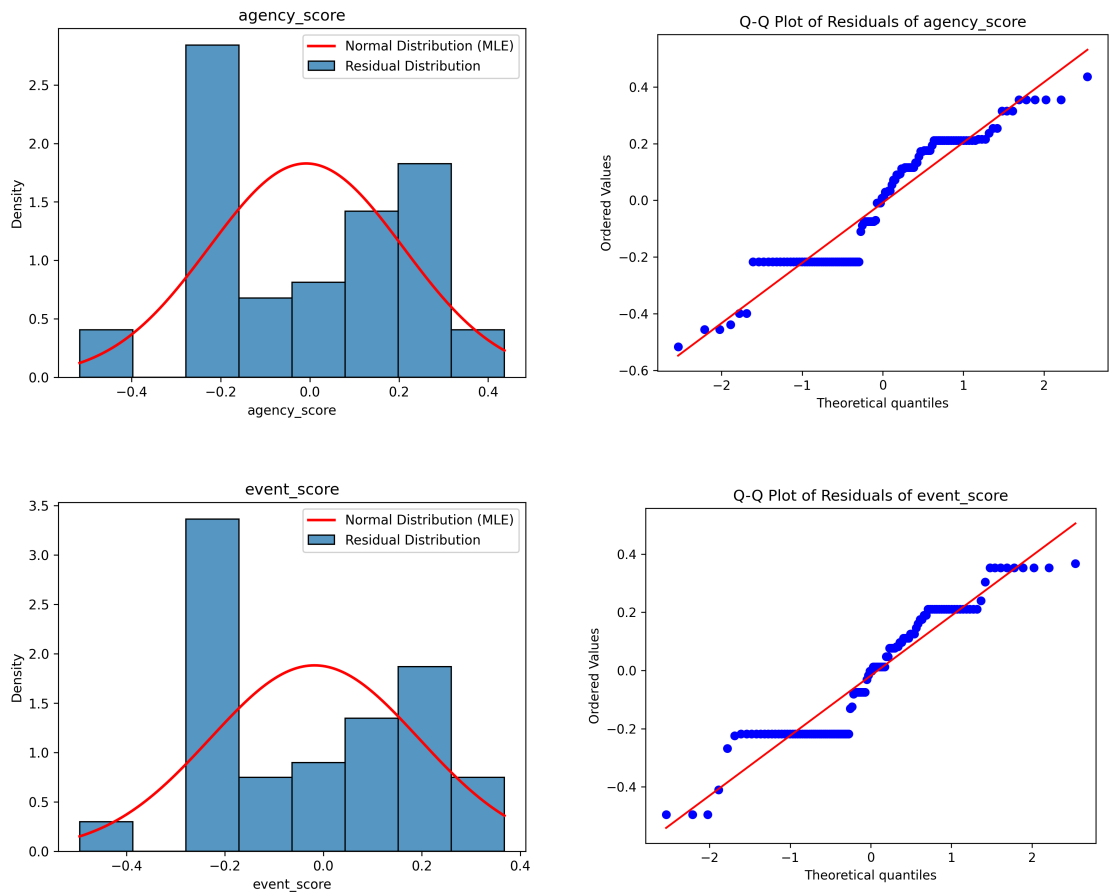
Linear regression and the coefficient of determination work building on the assumption that the residuals (actual values - predicted values) follow a normal distribution. A histogram of the residuals plotted alongside a maximum likelihood estimation distribution line gives a good indication of normality. Another way of visualizing normal distributions is using quantile-quantile (QQ) plots (Figure 3.). The x-axis represents the quantiles of the normal distribution and the y-axis the quantiles of the residuals. Therefore, if the dots in the

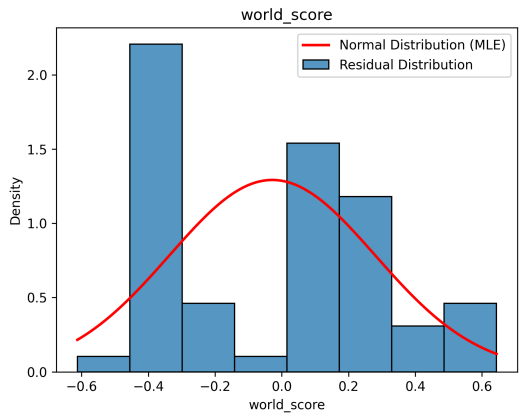
QQ plot align with the red line, the distribution is normal. *Curiosity, Suspense* and *Surprise* appear to have normally distributed residuals.

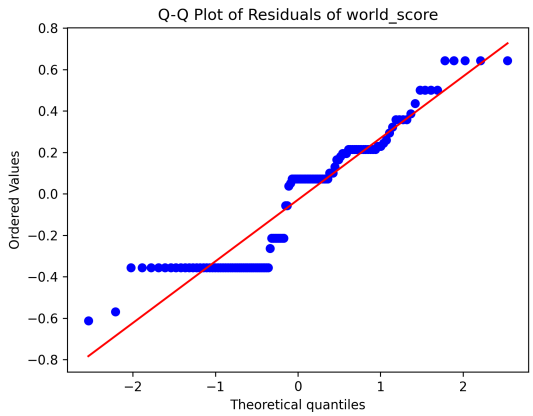
Table 1. *Linear Regression Results and Evaluation per Independent Variable*

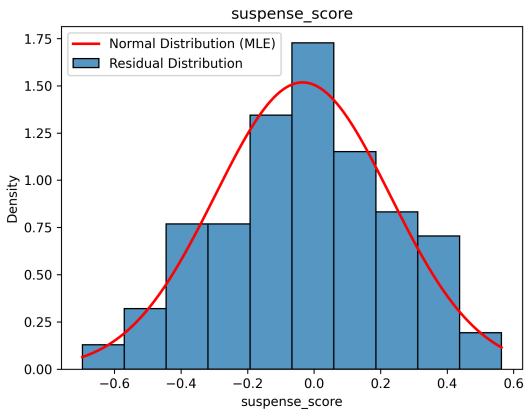
Independent Variable	**R^2**	**Slope**	**Intercept**
Agency	0.612	0.182	0.035
Event Sequencing	0.631	0.192	0.026
World Building	0.218	0.192	0.164
Suspense	0.428	0.260	-0.214
Surprise	0.344	0.238	-0.194
Curiosity	0.338	0.271	-0.323

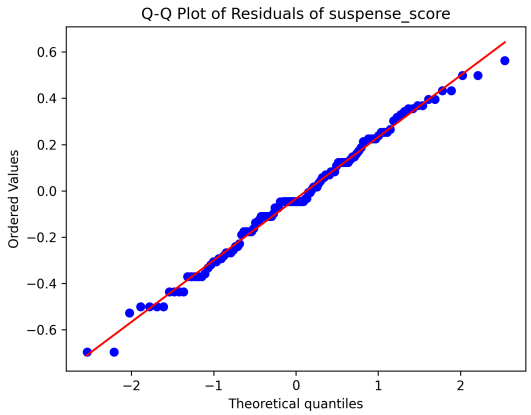
Figure 3. *Histograms of Residuals plotted against MLE distributions and Q-Q Plots*

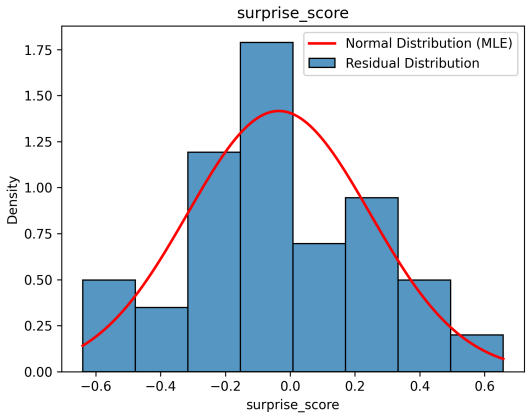


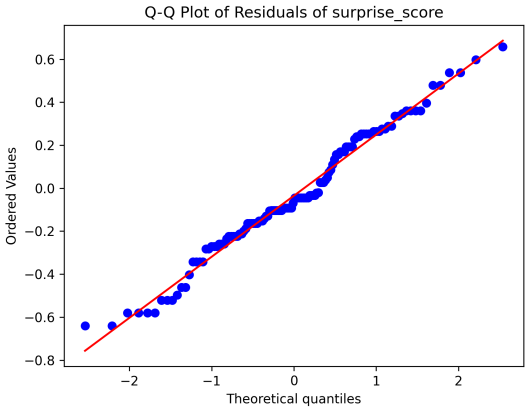

world_score
Normal Distribution (MLE)
Residual Distribution
Density
world_score

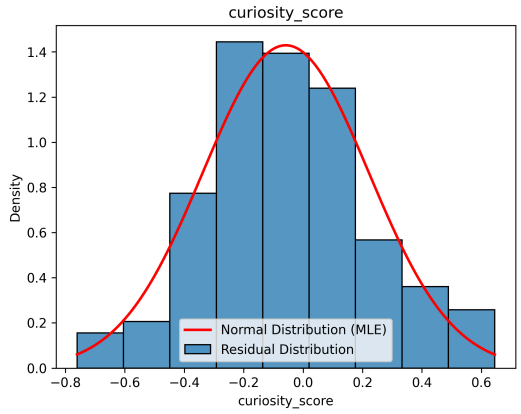

Q-Q Plot of Residuals of world_score
Ordered Values
Theoretical quantiles

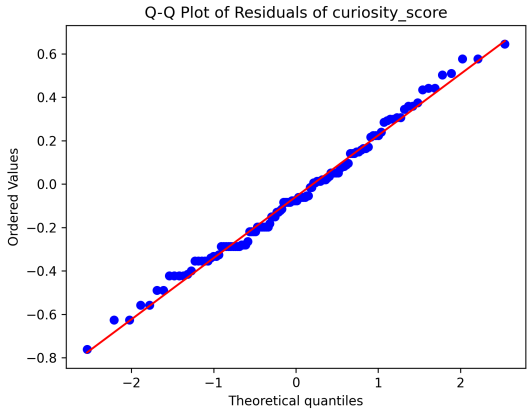

suspense_score
Normal Distribution (MLE)
Residual Distribution
Density
suspense_score


Q-Q Plot of Residuals of suspense_score
Ordered Values
Theoretical quantiles


surprise_score
Normal Distribution (MLE)
Residual Distribution
Density
surprise_score


Q-Q Plot of Residuals of surprise_score
Ordered Values
Theoretical quantiles


curiosity_score
Normal Distribution (MLE)
Residual Distribution
Density
curiosity_score


Q-Q Plot of Residuals of curiosity_score
Ordered Values
Theoretical quantiles

R^2 is more meaningful when the residuals can be considered statistically normally distributed, which is achieved by performing the Shapiro-Wilk test (Table 2.). Aligned with what we saw in Figure 3., *Suspense, Surprise* and *Curiosity* reject the Shapiro-Wilk null hypothesis that the data is not normally distributed. Therefore, we assume that the residuals of these independent variables have a normal distribution.

Table 2. *Shapiro-Wilk test on the Residuals of each Independent Variable*

Independent Variable	**p-value**
Agency	$p < 0.001$
Event Sequencing	$p < 0.001$
World Building	$p < 0.001$
Suspense	$p = 0.602$
Surprise	$p = 0.088$
Curiosity	$p = 0.527$

To conclude, *Agency* and *Event Sequencing* appear to be the best predictors of *Story Score*, but the residuals are not normally distributed ($p < 0.001$), which limits the credibility of the R^2 scores. *World Building* is not a very good predictor ($R^2 = 0.22$) and has no normal distribution of residuals, making it the worst predictor of *Story Score* among the independent variables. *Suspense, Surprise* and *Curiosity* possess normally distributed residuals, but lack the ability to explain the variance of *Story Score*, with *Suspense* having the highest score of the three ($R^2 = 0.43$). It seems no independent variable has a very meaningful correlation with the *Story Score*. Perhaps other factors such as the length of the stories, or maybe even the amount of upvotes on Reddit, are better predictors.